

Discrete Mathematics

CS1111

Lecture 2

Relations

Relations

- Let A and B be two sets, a relation R from A to B denoted $R: A \rightarrow B$ is a subset of the Cartesian product of A and B
 - $R: A \rightarrow B \subseteq A \times B$
- Clearly R is a set of ordered pair, such that the first element is from A and the second is from B

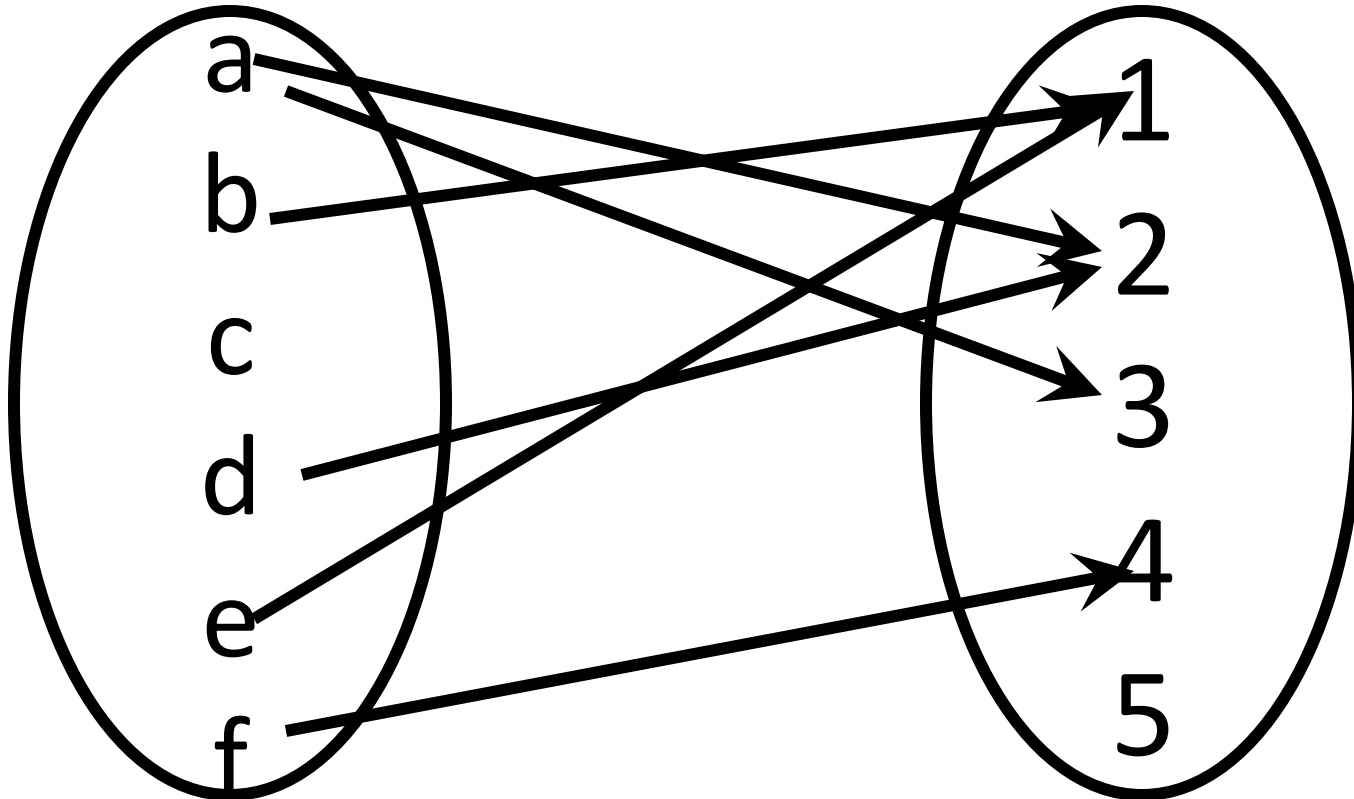
Example

- Let $X = \{\text{Ahmed, Ali, Omer, Salih}\}$,
 $Y = \{\text{Discrete Math, Programming}\}$

$R : X \rightarrow Y = \{(\text{Ahmed, Discrete Math}), (\text{Ali, Discrete Math}), (\text{Omer, Programming}), (\text{Salih, Programming}), (\text{Ali, Programming}), (\text{Salih, Discrete Math})\}$

Example

- Let $A = \{a, b, c, d, e, f\}$, $B = \{1, 2, 3, 4, 5\}$,
 $R:A \rightarrow B = \{(a, 3), (b, 1), (d, 2), (a, 2), (e, 1), (f, 4)\}$



Example

R	1	2	3	4	5
a		×	×		
b	×				
c					
d		×			
e	×				
f				×	

Some Terms

- Let $R: X \rightarrow Y$ defined by $R = \{(x, y) \mid x \in X \text{ and } y \in Y\}$
 - The set X is called the domain of the relation R
 - The set Y is called the co-domain of the relation R
 - For the ordered pair (x, y) y is the image of x and x is the pre-image of y .
 - The set of all images y is called the range of the relation. $\text{Range} \subseteq \text{co-domain}$

Relations on a Set

- A relation on a set A is a relation from A to A
- Let $A = \{a, b, c, d, e, f\}$

Then $R_1: A \rightarrow A = \{(a, a), (a, f), (b, e), (e, d), (c, f)\}$
is a relation on A

- Let $B = \{b \mid b \in \mathbb{N} \text{ and } b \leq 20\}$

Then $R_2: B \rightarrow B = \{(x, y) \mid x, y \in B \text{ and } y = 4x\}$ is
a relation on B

What are the members of R_2 ?

Reflexive Relations

- A relation R on a set A is said to be reflexive iff for all $x \in A$. $(x, x) \in R$
- Consider the following relations
 - $R_1: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \times y \in \mathbb{N} \}$
 - $R_2: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x + y \in \mathbb{N} \}$
 - $R_3: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \div y \in \mathbb{N} \}$
 - $R_4: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x < y \}$
 - $R_5: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \leq y \}$

Reflexive Relations

- $R_1: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \times y \in \mathbb{N} \}$
– $x \times x \in \mathbb{N} \rightarrow (x, x) \in R_1.$
- $R_2: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x + y \in \mathbb{N} \}$
– $x + x \in \mathbb{N} \rightarrow (x, x) \in R_2.$
- $R_3: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \div y \in \mathbb{N} \}$
– $x \div x = 1 \in \mathbb{N} \rightarrow (x, x) \in R_3.$

Reflexive Relations

- $R_4: N \rightarrow N = \{ (x, y) \mid x, y \in N \text{ and } x < y \}$
 - $x \text{ is NOT } < x \Rightarrow (x, x) \notin R_4.$
- $R_5: N \rightarrow N = \{ (x, y) \mid x, y \in N \text{ and } x \leq y \}$
 - $x \leq x \Rightarrow (x, x) \in R_5.$

Symmetric Relations

- A relation R on a set A is said to be symmetric iff for $a, b \in A$ if $(a, b) \in R$ then also $(b, a) \in R$
- Consider the following relations
 - $R_1: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \times y \in \mathbb{N} \}$
 - $R_2: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x + y \in \mathbb{N} \}$
 - $R_3: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \div y \in \mathbb{N} \}$
 - $R_4: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x < y \}$
 - $R_5: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \leq y \}$

Symmetric Relations

- $R_1: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \times y \in \mathbb{N} \}$
 - If $x \times y \in \mathbb{N}$ also $y \times x \in \mathbb{N}$, Symmetric
- $R_2: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x + y \in \mathbb{N} \}$
 - If $x + y \in \mathbb{N}$ also $y + x \in \mathbb{N}$, Symmetric
- $R_3: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \div y \in \mathbb{N} \}$
 - $(12, 3) \in R_3$ because 12 and $3 \in \mathbb{N}$ and $12/3 \in \mathbb{N}$ but $3/12 \notin \mathbb{N}$ therefore $(3, 12) \notin R_3$. , Not Symmetric

Symmetric Relations

- $R_4: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x < y \}$
 - $(x, y) \in R_4 \rightarrow x < y \rightarrow y \text{ is not } < x \rightarrow (y, x) \notin R_4$ not symmetric
- $R_5: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x \leq y \}$
 - $(5, 10) \in R_5 \rightarrow 5 \leq 10 \rightarrow 10 \text{ is not } \leq 5 \rightarrow (10, 5) \notin R_5$ not symmetric

Antisymmetric relation

- A relation R on a set A such that for all $a, b \in A$ if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$ is called antisymmetric
- Antisymmetric implies that no pair of distinct elements a and b with aRb and bRa

Examples

- Let $A = \{a, b, c, d\}$
 - $R_1 : A \rightarrow A = \{(a, b), (b, a), (c, d), (a, c)\}$
 - $R_2 : A \rightarrow A = \{(a, a), (a, b), (a, c), (c, d)\}$

Examples

- Let $A = \{a, b, c, d\}$
 - $R_1 : A \rightarrow A = \{(a, b), (b, a), (c, d), (a, c)\}$
 - $(a, a) \notin R_1 \rightarrow$ not reflexive
 - $(c, d) \in R_1$, but $(d, c) \notin R_1 \rightarrow$ not symmetric
 - (a, b) and (b, a) both $\in R_1 \rightarrow$ not anti-symmetric
 - $R_2 : A \rightarrow A = \{(a, a), (a, b), (a, c), (c, d)\}$
 - $(b, b) \notin R_2 \rightarrow$ not reflexive
 - $(a, b) \in R_2$, but $(b, a) \notin R_2 \rightarrow$ not symmetric
 - For all $x \neq y$, No (x, y) and (y, x) both $\in R_2 \rightarrow$ anti-symmetric

Examples

$$- R_3 : A \rightarrow A = \{(a, b), (b, c), (c, d)\}$$

$$- R_4 : A \rightarrow A = \{(a, a), (b, b), (c, c)\}$$

Examples

- $R_3 : A \rightarrow A = \{(a, b), (b, c), (c, d)\}$
 - $(a, a) \notin R_3 \rightarrow$ not reflexive
 - $(a, b) \in R_3$, but $(b, a) \notin R_3 \rightarrow$ not symmetric
 - For all $x \neq y$, No (x, y) and (y, x) both $\in R_3 \rightarrow$ anti-symmetric
- $R_4 : A \rightarrow A = \{(a, a), (b, b), (c, c)\}$
 - $(d, d) \notin R_4 \rightarrow$ not reflexive
 - $(a, a) \in R_4$, $(b, b) \in R_4$, $(c, c) \in R_4 \rightarrow$ symmetric
 - For all $x \neq y$, No (x, y) and (y, x) both $\in R_4 \rightarrow$ anti-symmetric

Examples

- $R_5 : \mathbb{N} \rightarrow \mathbb{N} = \{(x, y) \mid x < y\}$
 - $R_5 = \{(0,1), (0,2), (0,3) \dots (1, 2), (1, 3), (1, 4). \dots, (2, 3), (2, 4), (2, 5), \dots\}$

Examples

- $R_5 : \mathbb{N} \rightarrow \mathbb{N} = \{(x, y) \mid x < y\}$
 - $R_5 = \{(0,1), (0,2), (0,3) \dots (1, 2), (1, 3), (1, 4) \dots, (2, 3), (2, 4), (2, 5), \dots\}$
 - $(x, x) \notin R_5 \rightarrow$ Not reflexive
 - If $(x, y) \in R_5$ then $x < y \rightarrow y > x \rightarrow (y, x) \notin R_5. \rightarrow$ not symmetric
 - $\forall x, y \in \mathbb{N}$ and $x \neq y, (x, y) \in R_5 \rightarrow x < y \rightarrow y > x \rightarrow (y, x) \notin R_5 \rightarrow$ anti-symmetric

Examples

- $R_6 : \mathbb{N} \rightarrow \mathbb{N} = \{(x, y) \mid x \leq y\}$
 - $R_6 = \{(0,0), (0,1), (0,2) \dots (1, 1), (1, 2), (1, 3), \dots, (2, 2), (2, 3), (2, 4), \dots\}$

Examples

- $R_6 : \mathbb{N} \rightarrow \mathbb{N} = \{(x, y) \mid x \leq y\}$
 - $R_6 = \{(0,0), (0,1), (0,2) \dots (1, 1), (1, 2), (1, 3), \dots, (2, 2), (2, 3), (2, 4), \dots\}$
 - $\forall x \in \mathbb{N}, x \leq x \rightarrow (x, x) \in R_6 \rightarrow$ Reflexive
 - For all $x \in \mathbb{N}$, if $x \neq y$ If $(x, y) \in R_6$ then $x < y \rightarrow y > x \rightarrow (y, x) \notin R_6. \rightarrow$ not symmetric
 - For all $x, y \in \mathbb{N}$ and $x \neq y, (x, y) \in R_6 \rightarrow x < y \rightarrow y > x \rightarrow (y, x) \notin R_6 \rightarrow$ anti-symmetric

Transitive Relation

- A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$, for all a, b and $c \in A$.

Examples

- Consider the following relations
 - $R_1: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } y = 2 \times x \}$
 - $R_2: \mathbb{N} \rightarrow \mathbb{N} = \{ (x, y) \mid x, y \in \mathbb{N} \text{ and } x < y \}$

Examples

- R_3 is a relation on the set of integers
 - $R = \{(x, y) \mid x \text{ divides } y\}$ is it reflexive? Symmetric? Transitive?

Equivalence Relations

- Let A be a set and R be a relation on A , if R is reflexive, symmetric and transitive then it is said to be an equivalence relation
- Example: Let R be a relation on $\mathbb{N}$ defined by

$$R = \{(x, y) \mid x = y\}$$

Clearly for all $x \in \mathbb{N}$ $(x, x) \in R \Rightarrow$ reflexive

$(x, y) \in R \Rightarrow (y, x) \in R \Rightarrow R$ is symmetric

If $(x, y) \in R$ and $(y, z) \in R \Rightarrow x = y = z \Rightarrow (x, z) \in R \Rightarrow$
 R is transitive

– Therefore R is an equivalence relation

n-ary relations

- Let $A_1, A_2, A_3, \dots, A_n$ be n sets,

$$R : A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow \dots \rightarrow A_n =$$

$$\{(a_1, a_2, a_3, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, a_3 \in A_3, \dots, a_n \in A_n\}$$

$(a_1, a_2, a_3, \dots, a_n)$ is called an n -tuple and R is an n -ary relation

Inverse Relation

- Let $R: X \rightarrow Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}$
then the inverse of R denoted R^{-1} is defined
as
 - $R^{-1} = \{(y, x) \mid x \in X \text{ and } y \in Y\}$
- Let R_1 be a relation on N defined by
 - $R_1 = \{(m, n) \mid m = 2n\}$Then $R_1^{-1} = \{(m, n) \mid n = 2m\}$
or $R_1^{-1} = \{(n, m) \mid m = 2n\}$